# Alpha Beta Benchmark Design Methods Study

HELIX FORGE LAB / p04-full-001998 / RUN R184 / 2026-07-13

| ALPHA F1 | PRECISION | RECALL | BETA F1 |
|---|---|---|---|
| 0.873 | 0.881 | 0.865 | 0.846 |

## Decision context

This report records a bounded benchmark design evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

## Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

## Lineage

Upstream evidence: p04-full-000015 (models/model-cards/model-reviews/alpha-beta/calibration/tables/alpha-beta-calibration-metrics-v01.csv).

| Slice | Records | Precision | Recall | F1 |
|---|---|---|---|---|
| slice-01 | 171 | 0.878 | 0.870 | 0.874 |
| slice-02 | 202 | 0.895 | 0.883 | 0.889 |
| slice-03 | 189 | 0.877 | 0.859 | 0.868 |
| slice-04 | 248 | 0.894 | 0.872 | 0.883 |
| slice-05 | 249 | 0.876 | 0.848 | 0.862 |